

Arjuna JEE 2.0 2027

Maths

Basic Mathematics

DPP: 8

Total Duration - 40 Min

Q1 Solve for x :

- (A) $|x| > 5$
 (B) $|x - 1| \leq 3$
 (C) $|2x + 3| \geq 4$

Q2 Solve the following inequalities: $|5 - 2x| < 1$

Q3 Set of values of x satisfying $\frac{|x|-1}{|x|+2} > 0$ is

- (A) $(-\infty, -2) \cup (2, \infty)$
 (B) $(-2, -1) \cup (1, 2)$
 (C) $(-\infty, -1) \cup (1, \infty)$
 (D) $(-2, 2)$

Q4 Solution of inequality: $\frac{|x|+4}{|x|+3} \geq 0$

- (A) $x \in (-\infty, -4] \cup [-3, 3] \cup [4, \infty)$
 (B) $x \in (-\infty, -4] \cup (-3, 3) \cup [4, \infty)$
 (C) $x \in \mathbb{R}$
 (D) Not possible

Q5 The complete solution of inequality $0 < |3x + 1| < \frac{1}{3}$ is

- (A) $(-4/9, -2/9)$
 (B) $[-4/9, -2/9]$
 (C) $(-4/9, -2/9) - \{-1/3\}$
 (D) $[-4/9, -2/9] - \{-1/3\}$

Q6 The solution set for $7 \leq |x| + 4 < 10$

- (A) $x \in (-6, -3] \cup [3, \infty)$
 (B) $x \in (-6, 3] \cup [3, 6)$
 (C) $x \in (-6, -3] \cup [3, 6)$
 (D) $x \in (-\infty, -3] \cup [3, \infty)$

Q7 The complete solution set of $|2x + 3| > 5$ is

- (A) $(-\infty, -4)$
 (B) $(1, \infty)$
 (C) $(-\infty, -4) \cup (1, \infty)$
 (D) $(-\infty, -1) \cup (4, \infty)$

Q8 Solve the following inequalities: $\left| \frac{2}{x-4} \right| > 1$

Q9 Solution set of $|x + 1| + |x - 4| > 7$

- (A) $(-\infty, -2) \cup (5, \infty)$
 (B) $(-2, 5)$
 (C) $(-\infty, -1) \cup (6, \infty)$
 (D) None of these

Q10 The set of all real numbers x for which $x^2 - |x + 2| + x > 0$, is

- (A) $(-\infty, -2)$
 (B) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$
 (C) $(-\infty, -1) \cup (1, \infty)$
 (D) $(\sqrt{2}, \infty)$

Q11 Find solution set of values of x for inequation $||x - 1| + 2| \leq 4$

- (A) $[-1, 3]$
 (B) $(-\infty, -1) \cup (3, \infty)$
 (C) $(-\infty, -1)$
 (D) $(3, \infty)$

Q12 Solve the following inequalities:

$$|x + 2| - |x - 1| < x - \frac{3}{2}$$

Q13 Complete solution set of $||x + 1| - 1| - 1| \leq 1$, is

- (A) $x \in [2, 4]$
 (B) $x \in [-4, 2]$
 (C) $x \in [8, 10]$
 (D) $x \in [-6, 4]$

Q14 The complete set of values of x satisfying inequations $|x - 1| \leq 3$ and $|x - 1| \geq 1$, is

- (A) $[2, 4]$
 (B) $(-\infty, 2] \cup [4, \infty)$
 (C) $[-2, 0] \cup [2, 4]$
 (D) $[-2, 4]$

Q15 The set of all real x satisfying the inequality

$$\frac{3-|x|}{4-|x|} \geq 0, \text{ is}$$

- (A) $(-\infty, -4) \cup [-3, 3] \cup (4, \infty)$
 (B) $(-\infty, -4) \cup (4, \infty)$



(C) $(-\infty, -3) \cup (4, \infty)$

(D) $(-\infty, -3) \cup (3, \infty)$

Q16 If $x^2 - 5|x| - 14 < 0$, then sum of all its integral solutions is

Q17 The complete solution set of the inequation

$$2|x|^2 - 5|x| + 2 < 0, \text{ is}$$

(A) $(-2, 2)$

(B) $(-\frac{1}{2}, \frac{1}{2})$

(C) $(-2, -\frac{1}{2}) \cup (\frac{1}{2}, 2)$

(D) $(-\infty, -2) \cup (2, \infty)$

Q18 The complete solution set of inequality

$$\left| \frac{x^2 - 5x + 4}{x^2 - 4} \right| \leq 1 \text{ is } [\alpha, \beta] \cup [\gamma, \infty) \text{ then the value}$$

of $\alpha + 5\beta + 2\gamma$ is equal to



Answer Key

- Q1** A) $x \in (-\infty, -5) \cup (5, \infty)$
B) $x \in [-2, 4]$
C) $x \in (-\infty, -3.5] \cup [0.5, \infty)$

Q2 $x \in (2, 3)$

Q3 (C)

Q4 (C)

Q5 (C)

Q6 (C)

Q7 (C)

Q8 $x \in (2, 4) \cup (4, 6)$

Q9 (A)

Q10 (B)

Q11 (A)

Q12 $x \in (9/2, \infty)$

Q13 (B)

Q14 (C)

Q15 (A)

Q16 0

Q17 (C)

Q18 13



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